Byzantine Robustness Comparison: CNN vs. SNN Encodings (Complete Sweeps)

ByzFL SNN Benchmark

June 22, 2026

Contents

1 Introduction

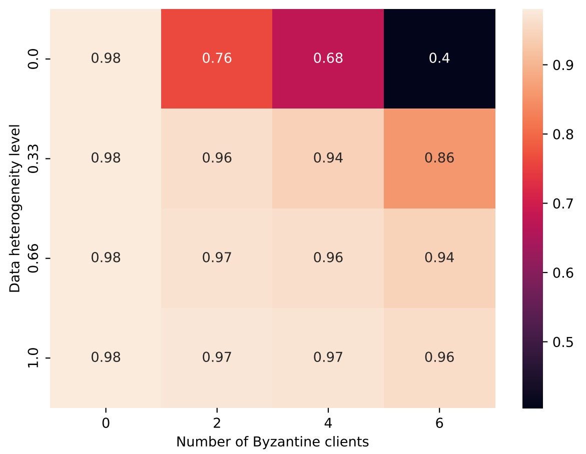
This document presents comparison grids (2x2 layout) of Byzantine robustness results for CNN and SNN architectures under different data encoding schemes. The configurations compared are:

1. **CNN**: Convolutional Neural Network on MNIST ('cnn_complete_plots')
2. **SNN Rate**: Spiking Neural Network with Rate encoding on MNIST ('snn_complete_plots_rate')
3. **SNN Direct**: Spiking Neural Network with Direct encoding on MNIST ('snn_complete_plots_direct')
4. **SNN NMNIST**: Spiking Neural Network on neuromorphic NMNIST ('snn_complete_plots_nmnist')

Each grid illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 6]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$).

2 Best Overall (Worst-Case Across All Attacks)

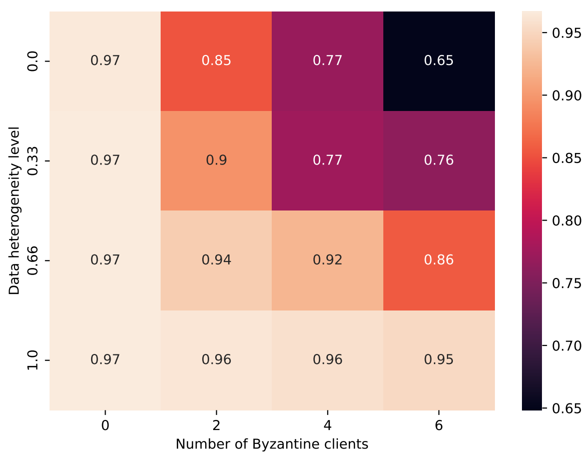
Figure 1 compares the best-performing aggregators under the worst-case attack scenario (minimizing the maximum damage over all attacks).



(a) CNN (Best Overall)

(b) SNN Rate (Best Overall)

(c) SNN Direct (Best Overall)

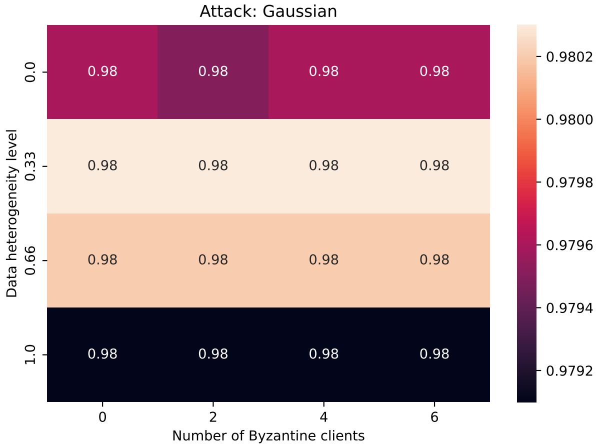


(d) SNN NMNIST (Best Overall)

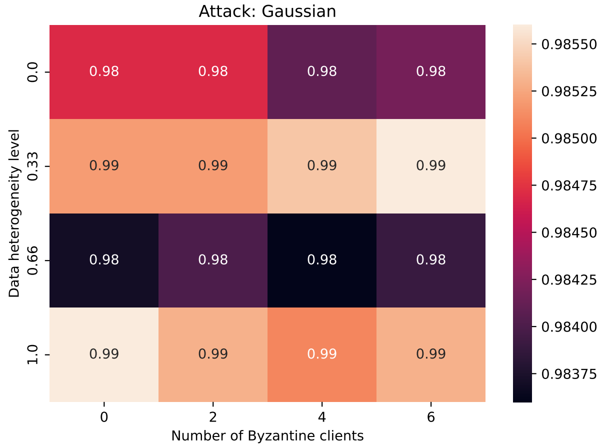
Figure 1: Best overall test accuracy comparison under worst-case attack across different data distribution parameters and number of Byzantine clients.

3 Fixed Attack: Gaussian Noise

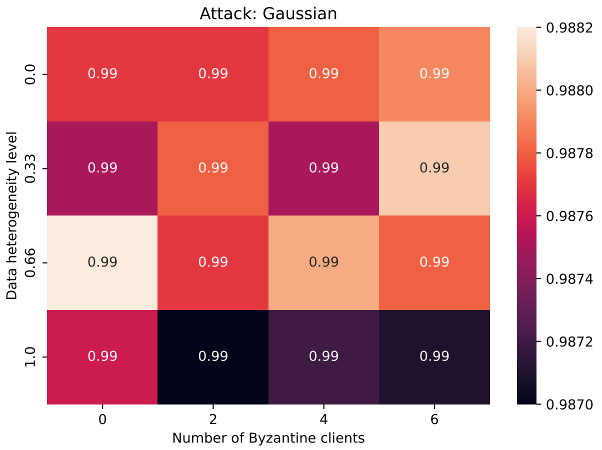
Figure 2 compares performance when the network is subjected to a Gaussian Byzantine attack.



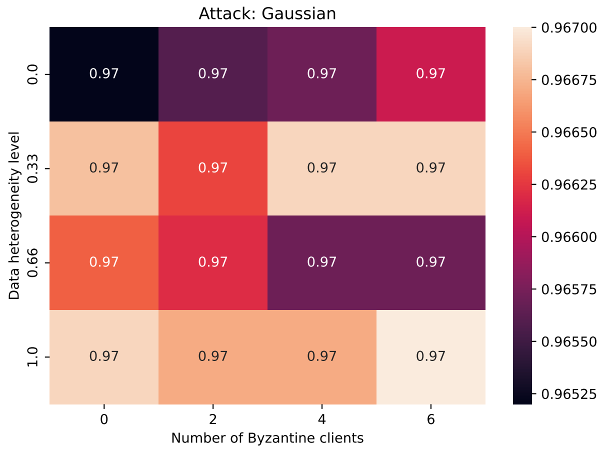
(a) CNN (Gaussian Attack)



(b) SNN Rate (Gaussian Attack)



(c) SNN Direct (Gaussian Attack)

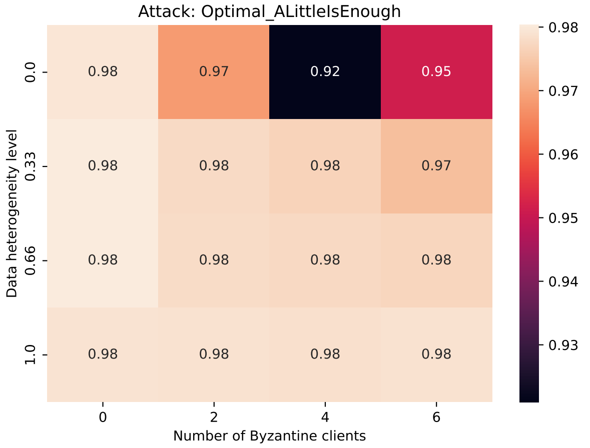


(d) SNN NMNIST (Gaussian Attack)

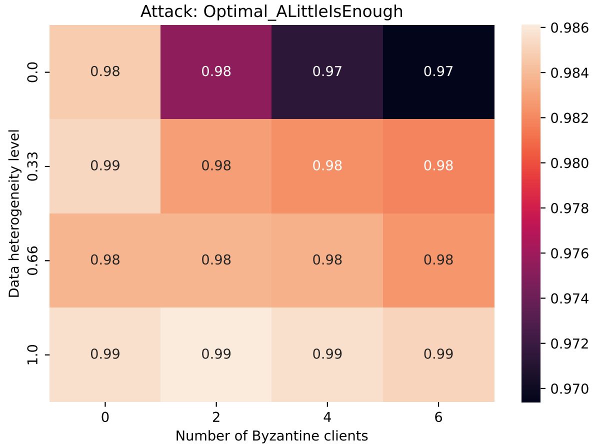
Figure 2: Accuracy comparison under Gaussian attack across different data distribution parameters and number of Byzantine clients.

4 Fixed Attack: Optimal A Little Is Enough (ALIE)

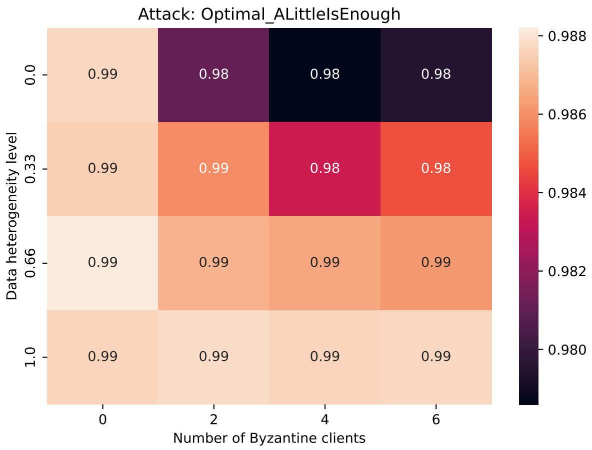
Figure 3 compares performance under the Optimal A Little Is Enough (ALIE) attack.



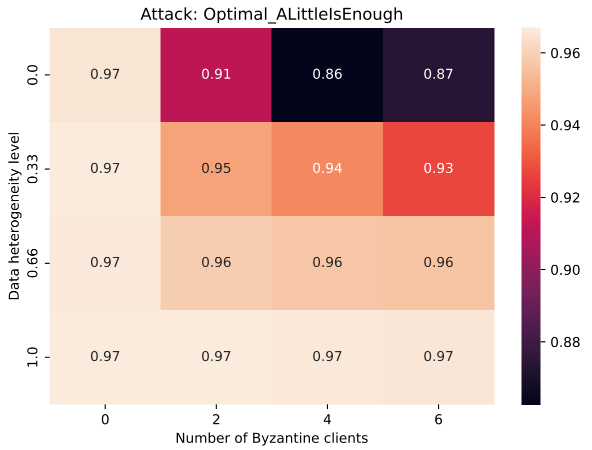
(a) CNN (ALIE Attack)



(b) SNN Rate (ALIE Attack)



(c) SNN Direct (ALIE Attack)

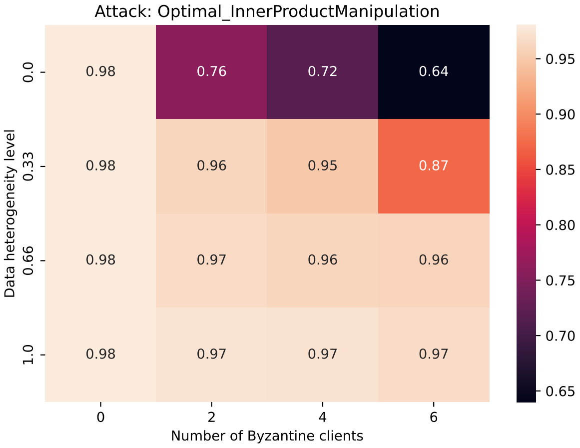


(d) SNN NMNIST (ALIE Attack)

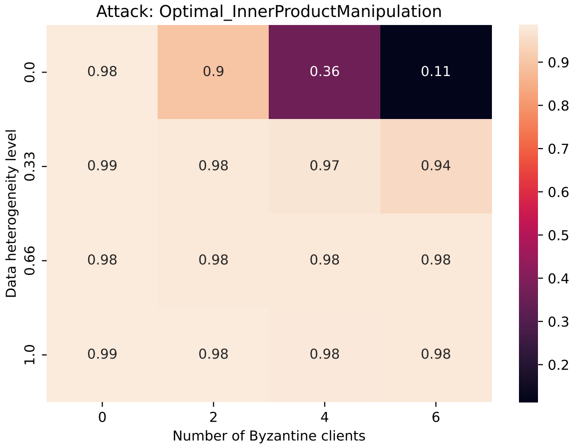
Figure 3: Accuracy comparison under Optimal ALIE attack across different data distribution parameters and number of Byzantine clients.

5 Fixed Attack: Optimal Inner Product Manipulation (IPM)

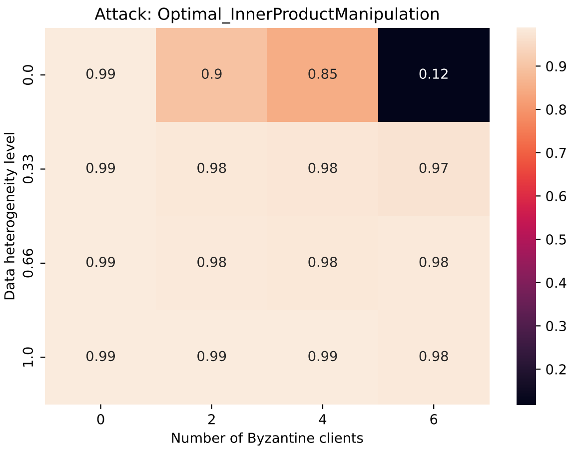
Figure 4 compares performance under the Optimal Inner Product Manipulation (IPM) attack.



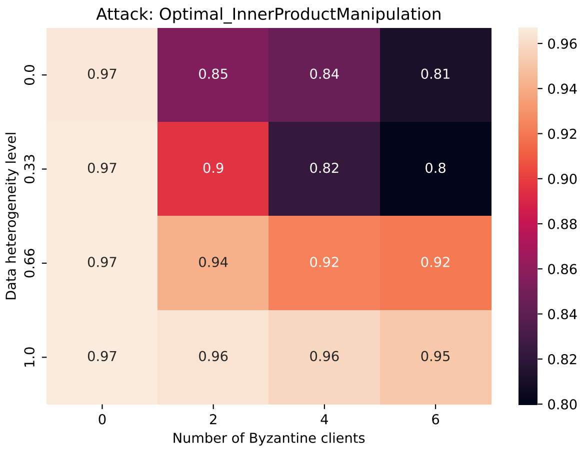
(a) CNN (IPM Attack)



(b) SNN Rate (IPM Attack)



(c) SNN Direct (IPM Attack)

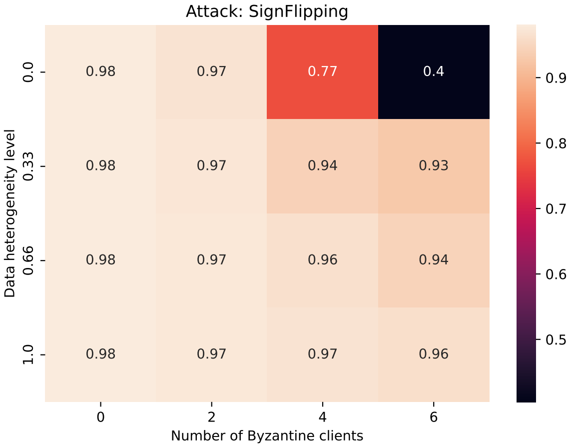


(d) SNN NMNIST (IPM Attack)

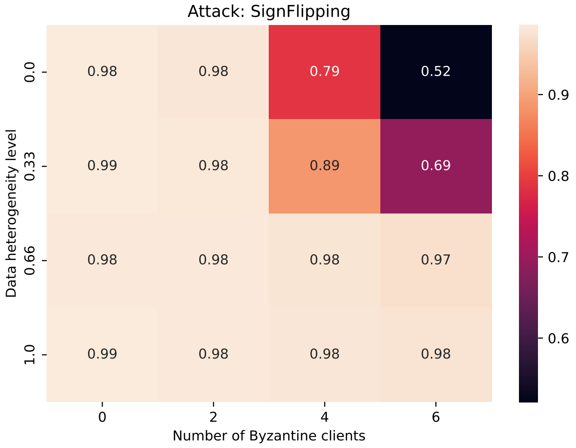
Figure 4: Accuracy comparison under Optimal IPM attack across different data distribution parameters and number of Byzantine clients.

6 Fixed Attack: Sign Flipping

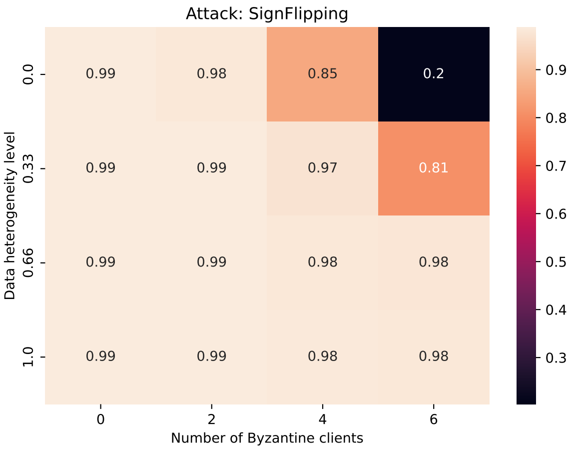
Figure 5 compares performance under the Sign Flipping attack.



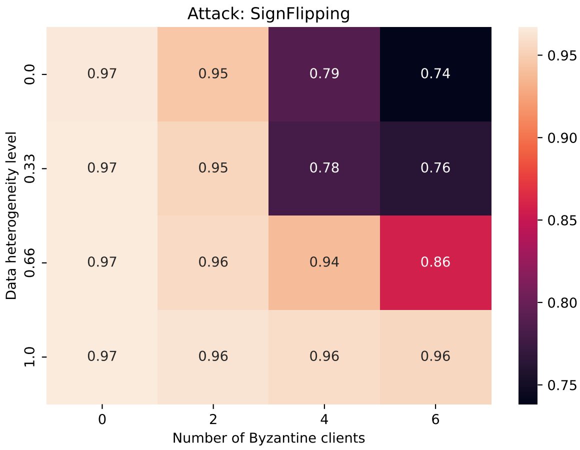
(a) CNN (Sign Flipping)



(b) SNN Rate (Sign Flipping)



(c) SNN Direct (Sign Flipping)



(d) SNN NMNIST (Sign Flipping)

Figure 5: Accuracy comparison under Sign Flipping attack across different data distribution parameters and number of Byzantine clients.